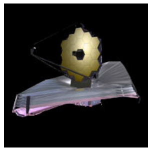

12182 - A Phase Resolved Study of Mass Transfer in Brown Dwarf Binaries

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	NIRSpec ZTFJ1239, G 140H/F100LP 1	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	2	NIRSpec ZTFJ1239, G 140H/F100LP 2	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	3	NIRSpec ZTFJ1239, G 140H/F100LP 3	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	4	NIRSpec ZTFJ1239, G 235H/F170LP 1	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	5	NIRSpec ZTFJ1239, G 395H/F290LP 1	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	6	NIRSpec ZTFJ1239, G 395H/F290LP 2	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347
	7	NIRSpec ZTFJ1239, G 395H/F290LP 3	NIRSpec Bright Object Time Series	(1) ZTFJ1239+8347

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	8	NIRSpec ZTFJ1444, G 140H/F100LP 1	NIRSpec Bright Object Time Series	(2) ZTFJ1444+4820
	9	NIRSpec ZTFJ1444, G 140H/F100LP 2	NIRSpec Bright Object Time Series	(2) ZTFJ1444+4820
	10	NIRSpec ZTFJ1444, G 140H/F100LP 3	NIRSpec Bright Object Time Series	(2) ZTFJ1444+4820
	11	NIRSpec ZTFJ1444, G 235H/F170LP	NIRSpec Bright Object Time Series	(2) ZTFJ1444+4820
	12	NIRSpec ZTFJ1444, G 395H/F290LP	NIRSpec Bright Object Time Series	(2) ZTFJ1444+4820

ABSTRACT

We propose near-infrared phase-resolved spectroscopy of two objects from a new, rare type of system --- accreting brown dwarf binaries (aBDBs) --- to obtain critical measurements only accessible with JWST. We will observe with NIRSpec across the near-infrared to do detailed, phase-resolved atmospheric retrieval, which will allow us to study these systems' compositions and determine the masses of the brown dwarf (BD) components. Mass transfer between substellar objects has not been seen before, and our two systems are the first of their kind. We detected two strong candidates, ZTF J1239+8347 and ZTF J1444+4820 through periodic variability searches in public Zwicky Transient Facility (ZTF) data. These systems exhibit extreme optical and UV variability with short periods around an hour. This behavior is typically only found in systems with compact objects. Since the photometric variability peaks in ultraviolet, the binary components' atmospheres are detectable in the near-infrared spectrum at all times. This proposal is designed to efficiently obtain all available near-infrared spectral information on these two new systems and characterize their atmospheres and the accretion physics at work in full detail.

OBSERVING DESCRIPTION

In order to constrain the compositions and masses of both the donor and accretor stars in these new accreting brown dwarf binaries (aBDBs), ZTFJ1239+8347 and ZTFJ1444+4820 we will observe both systems with JWST/NIRSpec in the high resolving power R~2700 mode across the near-infrared spectrum. We will obtain 6 minute exposures across three periods between 0.97-1.82 microns where the system atmospheres are brightest, and one period each in the two bands between 1.66-3.05 and 2.87-5.14 microns to fully characterize the systems compositions.

We will derive radial velocity (RV) measurements of both components in every epoch to obtain dynamical masses of each component in both systems. This, in combination with precise temperatures, will allow us to determine the mass transfer rate, accretion effiiciency, age, and stability timescale of the systems.

We optimized our exposure patterns using the JWST ETC with Phoenix low mass stellar atmosphere models to ensure we would obtain high enough SNR measurements to measure abundances accurately. Fiducial RVs of the accretors computed with optical spectra of the accretion hotspots confirm that the R~2700 resolving power of JWST/NIRSpec will definitively detect the RV signatures of both components in an SB2. Theoretical mass-period relations confirm that both components will be bright enough to contribute significantly to the near-infrared spectra.

We will observe contiguously in each filter to obtain a time series of observations for each target which will allow us to detect asymmetries in composition between the accretion hotspots and the general atmosphere of the accretor, as well as the roche lobe overflow pointa of the donors.

The specific time of observation is allowed to float for schedulability and is not relevant to the observations.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Multiple stars, Periodic variable stars] Extended=NO				
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	
Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Star Description=[Brown dwarfs, Multiple stars, Periodic variable stars] Extended=NO				

Observation

Proposal 12182, Observation 1: NIRSpec ZTFJ1239, G140H/F100LP 1

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 2: NIRSpec ZTFJ1239, G140H/F100LP 2

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 3: NIRSpec ZTFJ1239, G140H/F100LP 3

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 4: NIRSpec ZTFJ1239, G235H/F170LP 1

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 5: NIRSpec ZTFJ1239, G395H/F290LP 1 Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 6: NIRSpec ZTFJ1239, G395H/F290LP 2

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements	
	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 7: NIRSpec ZTFJ1239, G395H/F290LP 3

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1239+8347	RA: 12 39 55.3822 (189.9807592d) Dec: +83 47 7.52 (83.78542d) Equinox: J2000	Proper Motion RA: -22.654 mas/yr Proper Motion Dec: -19.794 mas/yr Parallax: 0.0029947" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSRAPID	198	13	1	1	13	2333.74	295104

Special Requirements	Time Series Observation No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Observation

Proposal 12182, Observation 8: NIRSpec ZTFJ1444, G140H/F100LP 1

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements

Time Series Observation
No Parallel Attachments

Group Observations 8, 9, 10, 11, 12, Non-interruptible

Observation

Proposal 12182, Observation 9: NIRSpec ZTFJ1444, G140H/F100LP 2

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements

Time Series Observation
No Parallel Attachments

Group Observations 8, 9, 10, 11, 12, Non-interruptible

Observation

Proposal 12182, Observation 10: NIRSpec ZTFJ1444, G140H/F100LP 3

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	198	20	1	1	20	3590.37	295104

Special Requirements

Time Series Observation
No Parallel Attachments

Group Observations 8, 9, 10, 11, 12, Non-interruptible

Observation

Proposal 12182, Observation 11: NIRSpec ZTFJ1444, G235H/F170LP

Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSRAPID	198	20	1	1	20	3590.37	

Special Requirements

Time Series Observation
No Parallel Attachments

Group Observations 8, 9, 10, 11, 12, Non-interruptible

Observation

Proposal 12182, Observation 12: NIRSpec ZTFJ1444, G395H/F290LP Thu May 21 13:00:28 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	ZTFJ1444+4820	RA: 14 44 9.3262 (221.0388592d) Dec: +48 20 44.33 (48.34565d) Equinox: J2000	Proper Motion RA: -25.116 mas/yr Proper Motion Dec: 11.339 mas/yr Parallax: 0.00150" Epoch of Position: 2016	

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Star
Description=[Brown dwarfs, Multiple stars, Periodic variable stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	295104

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSRAPID	198	20	1	1	20	3590.37	

Special Requirements

Time Series Observation
No Parallel Attachments

Group Observations 8, 9, 10, 11, 12, Non-interruptible